INCH-POUND

MIL-L-70830(AR)

25 August 1989

MILITARY SPECIFICATION

LEAD, HNS TEST, ASSEMBLY AND PACKAGING FOR

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification contains the requirements and quality assurance provisions for fabrication of parts, assembly and packing for a lead assembly.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-109	-Quality Assurance Terms and Definitions.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-1168	-Lot Numbering of Ammunition.
MIL-STD-1169	-Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1190	-Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copes of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT

9384813	-Lead Assembly.

PACKAGING

7548101	-Box, Packing Ammo for Primer and Delay Assembly.
7548103	-Box, Set-Up Packing, Ammo Outer for Small Explosive Components w/o Primary Explosives.
7548104	-Box, Set-Up Packing, Ammo Inner, for Small Explosive Components w/o Primary Explosives.

INSPECTION

12913007	-Lead Propagation Test Assembly

(Copies of drawings and publications required by manufacturers in connection with specific acquisiion functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications. (See 4.5.1.)

3.2 Components and assemblies. The components and assemblies to be used in first article samples and production lots shall comply with all requirements specified on Drawing 9384813 and associated drawings and with all requirements specified in applicable specifications and standards. (See 4.5.1.)

3.3 Manufacturing process. All parts and assemblies shall be manufactured by the equivalent process used for first article approval and no deviations from that process shall be made without contracting officer approval.

3.4 Function.

3.4.1 Sensitivity and output. The lead assembly when initiated by a high order detonator output will function high order and will indent the steel witness block to a minimum depth of 0.020 inch when tested as specified in 4.5.2.1.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.3).

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil, corrosion products and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface shall be continuous except for a few light scratches not exposing base material. All required markings and stampings shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 and MIL-A-48078 shall apply.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection (See 4.3).

b. Quality Conformance Inspection (See 4.4).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the requirements and test method paragraphs in this specification.

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Lead Assemblies	9384813	315 (min.)

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4.3.2 Inspections to be performed. The examinations and tests listed in Table I shall be performed on the first article sample (see MIL-A-48078).

4.3.3 Rejection. If any of the applicable components, subassemblies or assembly fails to comply with the requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon the failure of lead components, subassembly or assembly to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Lead Assembly	1 OF 1	9384813
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Lead Assembly			
	Examination for Defects Functioning - Sensitivity and Output	315 2 - 3 315 0 - 1, Level II	3.2.3.3 3.4.1	4.4.2.1 and 4.4.2.2 4.5.2.1

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawing, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot number, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain PBXN-5 from an acceptable lot from one manufacturer with one lot interfix number.

4.4.2 Examinatins and tests. The examinations listed herein shall be performed on inspection lots as defined in 4.4.1. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

a. Classificaiton of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

b. Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the

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proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

c. Whenever possible, utilize non-operator dependent test equipment. Accepted items should be marked accordingly. Where specified herein, attributes sampling inspection shall be conducted in accordance with Table II below. Samples should be selected at random.

Table II below contains a selection of 0-1 plans, indexed by Inspection Levels and Lot Size. The Inspection Level indices are new, but can be related to the standard in that comparable consumer risk levels are maintained. Columns II and III are recommended for most major characteristics; columns IV and V for minors. Such usage would be consistent with past and current spec practices. Column I is intended for use where there must be extremely low risk of accepting nonconforming product. Column VI should only be used where a high risk of accepting nonconforming product must be tolerated.

TABLE II.
0-1 ATTRIBUTES PLANS

Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10000 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

* 100% = inspection

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Lead Assembly (Non-destructive Test)	1 OF 2	9384813
			NEXT HIGHER ASSEMBLY 11820609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MAJOR				
101	Total height	0-1, Level II	3.2	SMTE
102	Outside diameter	0-1, Level II	3.2	SMTE
103	Cracks or splits	100%	3.2	Visual
104	Caked explosives on exterior	100%	3.2	Visual
MINOR				
201	Crimp not full 360°	0-1, Level IV	3.2	Visual
202	Stencil, ink, anodize, missing or wrong color	0-1, Level IV	3.2	Visual
203	Poor workmanship other than critical above	0-1, Level IV	3.6	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-L-70830(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9384813
4.4.2.2	Lead Assembly (Destructive Tests)		NEXT HIGHER ASSEMBLY 11820609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101	Sensitivity and output	0-1, Level II	3.4.1	4.5.2.1
MINOR	None defined			

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4.3 Testing. The tests (first article and quality conformance) shall be performed on inspection lots as defined in 4.4.1. The sampling plans for test shall conform to the provisions of 4.4.2.c.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column starting with 4.4.2.1. The contractor shall use inspection equipment designs in accordance drawing 12913007. See Section 6.2 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of chemical, toxic explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Functioning.

4.5.2.1 Sensitivity and output. A random sample of leads shall be selected from each lot. The lead shall be assembled with a detonator (Dwg 9384806) into the test setup specified in Dwg. 28102214, a 0.25 ± 0.01 ounce weight shall be dropped on the firing pin from a height of four inches, maximum. A lead that fails to fire or fires but fails to produce a dent in the steel witness block of the required depth shall be classed defective if the detonator confinement sleeve expansion is 0.330 inch diameter or greater. If the confinement sleeve expansion is less than 0.330 inch diameter, the lead assembly shall not be classed defective, but the detonator shall be classed defective and the result a no-test. The indentation depth shall be measured using a probe with a 0.005 inch + 0.017 inch spherical radius tip.

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5. PACKAGING

5.1 Unit packaging.

5.1.1 Level A. Not available

5.1.2 Level B. Unit packaging for the Lead Assembly shall be in accordance with Dwg 7548104 SH2 Line 8.

5.1.3 Level C. Same as level B.

5.2 Packaging.

5.2.1 Level A. Not available

5.2.2 Level B. Packaging for the Lead Assembly shall be in accordance with Dwg 7548103 SH2 Line 8.

5.2.3 Level C. Same as level B.

5.3 Packing.

5.3.1 Level A. Not available

5.3.2 Level B. Packing for the Lead Assembly shall be in accordance with Dwg 7548101 SH3.

5.3.3 Level C. Same as level B.

5.4 Marking.

5.4.1 Level A. Not available

5.4.2 Level B. Unit packaging, packaging and packing shall be in accordance with Dwgs. 7548103, 7548104 and 7548101.

5.4.3 Level C. Same as level B.

5.5 Packaging, packing and marking for inert lead assembly parts.

5.5.1 Packaging packing and marking. Inert lead parts shall be in accordance with MIL-STD-1169 for all applicable levels of packaging.

5.6 Shipping. When Lead from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

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5.6.1 Unitization. Unitization shall be in accordance with MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lead assemblies are intended for use in the following fuze:

25mm	M758	11820609

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.

b. Place of inspection if not at the place of manufacture.

c. Instructions for shipment of production lot samples for functional tests.

d. Provision for preparation for delivery if required for indefinite storage of inter-plant shipment of the lead assemblies.

e. Requirement for contractor to provide and maintain an inspection system in accordance with MIL-Q-9858, Quality Program Requirements.

f. Provisions for MIL-STD-1167, Ammunition Data Cards on DD Form 1423, Contract Data Requirements List.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, ARDEC ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Check test for possible deterioration. The following test is designed to prevent leads that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the leads are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

6.4.1 Functioning. The sample size and acceptance or rejection criteria shall be in accordance with Table II. Test shall be performed as specified in 4.5.3, using equipment in accordance with, 4.4.4.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Subject term (keyword) listing.

Lead assembly
M758 Fuze
25mm

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A670)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-70830 (AR)	25 August 1989

3. DOCUMENT TITLE

LEAD, HNS TEST, ASSEMBLY AND PACKAGING FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Specifications & Standardization Office	(201) 724-6671	880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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